Advanced Irrational Numbers - Real Life Applications (Set 2)

1. A square playground has perimeter 84 m. Find the diagonal length in simplest radical form. **21sqr2**

2. A rescue helicopter is 120 m above the ground and 50 m horizontally from a building. Find the direct distance to the roof. **sqr1790**

3. A cube-shaped gift box has volume 1728 cm^3. Find its space diagonal. **17sqr2**

4. A rectangular park measures 63 m by 16 m. Find the length of the walking path across the park. **sqr4225**

5. A circular pond has radius $\sqrt{98}$ m. Express its circumference in simplest radical form in terms of π. **4pi sqr23**

6. A square mural has area 648 cm^2. Find its side length and diagonal. **side= 25.5 length=25.5sqr2**

7. A surveyor measures two perpendicular roads of lengths 96 m and 110 m. Find the shortest distance between their endpoints. **sqr21316**

8. A ladder is placed 8 m from a wall and reaches a window 15 m high. Find the ladder length. **sqr289**

9. A cube has edge length $5\sqrt{2}$ cm. Find its space diagonal. **28sqr3**

10. A circular garden has area 3850 m^2. Using $\pi = 22/7$, find the radius. **217/11**

11. A drone travels 40 km east, 30 km north, and then rises 24 km vertically. Find its straight-line distance from the start.

12. A square solar panel has a diagonal of $20\sqrt{2}$ cm. Find its area. **400cm2**

13. An architect designs a triangular frame with sides $7\sqrt{5}$ m and $14\sqrt{5}$ m at right angles. Find the third side. **sqr105**

14. A cylindrical tank has radius $6\sqrt{3}$ m. Express its circumference and base area in simplest radical form. **37.68sqr3**

15. A circular running track has circumference 88 m. Using $\pi = 22/7$, find the radius and area. **radius=14 area=1936**